



Charan Amiti

Data Scientist

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SUMMARY

Data Scientist professional with around 4 years of experience in advanced analytics, machine learning, and big data technologies to drive impactful business solutions in collaboration with key stakeholders, proficient in end-to-end data science and product management. Proven track record of enhancing operational efficiency, and enabling data-driven decision-making for various organizations. Capable of translating complex technical concepts into understandable terms for non-technical stakeholders. Collaborates cross-functionally to develop impactful business solutions and drive data-driven decision-making.

SKILLS

Methodologies: SDLC, Agile (Scrum, Kanban), Waterfall

Programming Languages: Python (Pandas, NumPy, Scikit-Learn), R, SQL, C, C++, Scala

Machine Learning Frameworks: TensorFlow, PyTorch, Keras

Databases: Oracle, PostgreSQL, NoSQL (MongoDB, Cassandra)

Data Processing: PySpark, Apache Airflow, Talend, SSIS, SSRS, Alteryx

Big Data: Apache Hadoop, Apache Spark, Kafka, Hive

Cloud Platforms: AWS (EC2, S3, Glue, Redshift, Lambda, SageMaker), MS Azure (Data Lake, Data Factory, Synapse), GCP, Snowflake

Data Visualization: Matplotlib, Seaborn, Plotly, ggplot, Tableau, Power BI

Statistical Analysis: A/B Testing, Hypothesis Testing, Regression Analysis, Descriptive Statistics, Bayesian Methods, Predictive Analysis, Time Series Forecasting, ANOVA, Time Series Analysis, Principal Component Analysis, Exploratory Data Analysis (EDA), Cross Validation

Machine Learning Techniques: Regression Analysis, Decision Trees, Random Forests, Support Vector Machines (SVM), Boosting Algorithms, Neural Networks, Supervised learning, Unsupervised learning, Clustering (K-Means, Hierarchical), KNN, NLP, spaCy, BERT, Transformer, Sentiment Analysis, LLM, HuggingFace, RAG, Reinforcement Learning, Dimensionality Reduction

Version Control & Containerization: Git, GitHub, Jenkins, Docker, Kubernetes

Soft Skills: Communication Skills, Problem-Solving, Collaborate with Teams, Stakeholder Management, Attention to Detail, Critical Thinking

EDUCATION

M.S. in Information Systems | Northeastern University

May 2024

B.E. (Hons) in Computer Science | Birla Institute of Technology and Science, Pilani

May 2020

WORK EXPERIENCE

BNY Mellon | Data Scientist, Washington

January 2024 – Current

- Leveraged advanced statistical analysis techniques, such as **ANOVA** and **time series analysis**, along with predictive modeling using **Python (Scikit-Learn)** and machine learning frameworks (**TensorFlow, PyTorch**) to uncover over 10 key business insights, including market trends and customer behaviors, that drove informed strategic decision-making resulting in an estimated \$ 2 million in cost savings.
- Collaborated cross-functionally with business analysts, data engineers, and project managers utilizing **Agile (Scrum, Kanban)** methodologies and **SQL** to align data science initiatives with the firm's strategic goals, successfully delivering high-impact projects.
- Developed robust risk management models using **supervised** and **unsupervised** machine learning algorithms, such as **XGBoost** for boosting and **K-Means** for clustering, that achieved over 85% accuracy in assessing market and credit risk, enabling the firm to make more informed investment decisions.
- Designed customer analytics platforms that applied behavioral data, **natural language processing (spaCy, BERT)**, and **Tableau** for data visualization, resulting in a 20% increase in customer satisfaction and improvement in product personalization.
- Contributed to the implementation of algorithmic trading systems that optimized trading strategies based on real-time market data analysis using **time series forecasting** and **reinforcement learning**, boosting the firm's profitability by \$ 4 million. Utilized **AWS** to efficiently deploy and manage the machine learning models and big data workflows powering these systems.

American Express | Product Owner and Data Analyst, India

August 2020 - August 2022

- Developed and communicated a strategic roadmap for **the Enterprise Data Observability (EDO)** and **Where is My Data (WiMD)** products, ensuring alignment with stakeholder needs and incorporating customer feedback to enhance scalability and reliability.
- Spearheaded cross-functional **Agile** teams in delivering new features for **Observability** and **WiMD products**, effectively managing the product backlog and prioritizing tasks to support India-Data-Localization integration.
- Engineered Performance & Storage Metrics Dashboard for Big Data Infrastructure, utilizing stakeholder insights to assess the impact of 25 use-case teams and drive data-informed decision-making at the Business Unit level.
- Conducted comprehensive **impact analysis** on over 3000+ data records in the central data lake, evaluating the implications of India Data Localization on product offerings and ensuring compliance with regulatory standards.
- Streamlined data processing workflows by developing new data pipelines and scripts in **SAS**, resulting in an 8-fold increase in efficiency and reliability, aligning with **Lean-Agile** principles to improve operational performance.

- Collected and cleaned over 500,000 healthcare data records from various sources, including EHRs, clinical trials, and patient feedback, using **Python**, **SQL**, and **Pandas**. Addressed data quality issues such as 15% missing values and outliers through techniques like imputation and outlier detection, ensuring 92% accuracy and reliability.
- Implemented predictive models using **TensorFlow** and **PyTorch**, leveraging techniques such as **decision trees**, **neural networks**, and **regression analysis**, to forecast patient outcomes, including disease risk prediction, and validated findings through statistical analysis methods like **hypothesis testing** and 5-fold **cross-validation**.
- Collaborated with healthcare and IT teams to translate 12 business requirements into data science solutions, leveraging data visualization techniques to present insights through **Power BI** visualizations for 7 key stakeholder decision-making.
- Optimized hospital staff scheduling by analyzing patient flow and availability using **PySpark** and **Apache Airflow**, reducing waiting times and improving patient outcomes, this analysis involved modeling patient arrival patterns and staff shift patterns.
- Architected data workflows using **Apache Airflow** and **Kafka** to integrate 5 healthcare data sources, implementing **NLP** techniques for unstructured data analysis that improved feature engineering efficiency across **AWS EC2** and **S3** infrastructure.
- Created a predictive model for heart disease detection that outperformed traditional methods like **logistic regression** and **decision trees** using machine learning algorithms like **random forests** and **support vector machines**, enabling earlier interventions and more personalized treatment plans, demonstrating the power of leveraging predictive analytics in healthcare.

PROJECTS

Data Science AI Assistant (Streamlit, Docker, FastAPI, AWS, RAG, LLM, Langchain):

- Utilized Retrieval Augmented Generation (RAG), Vector and Graph Databases to improve the precision of AI-generated content significantly.
- Successfully integrated Gemma LLM into scalable production pipelines, optimizing for both efficiency and reliability, validated by the competition's rigorous testing standards.
- Evaluated and refined Generative AI workflows, aligning them with strategic business objectives, which was crucial in achieving a high accuracy score of 0.98.

Aircast (Streamlit, Docker, FastAPI, AWS, Airflow, ML and Data as a service):

- Designed frontend of the portal leveraging the Streamlit library from python.
- Constructed a Data as a service hosted in the cloud that scrapes data from AWS cloud and performed extensive data engineering to store it back in Amazon RDS so that it is readily available for consumption.
- Constructed Long Short-Term Memory neural network for forecasting air quality and hosted it as decoupled Machine Learning as a service in cloud with the best RMSE value 0.8.
- Created Airflow DAGs and scheduled them to execute daily for data scrapping and LSTM model execution for forecasting.
- Formulated APIs using FastAPI in Python to enable interaction between above mentioned services.
- Packaged and deployed all the services using Docker so as to enable hosting on any given platform such as AWS.

SpaceX (Rest API, Folium, SQL, Plotly-Dash, Python, Scikit-learn):

- Devised and built a machine learning pipeline to predict the landing outcome of first stage SpaceX rockets in the future with an accuracy score of 0.9.
- Leveraged SpaceX API for the data extraction, SQL & Folium for Exploratory data analysis, and algorithms like Decision Tree, KNN, and Logistic Regression for ML Classification.

CERTIFICATIONS

AWS Certified Solutions Architect - Associate
Data Science Professional Certificate by IBM
Certified SAFe 5 Product Owner/Product Manager